GradeOS Simulator

v0.7 User Guide

Comprehensive Instruction Manual

RTK GPS Machine Control Simulator

Dozers • Excavators • Terrain Grading

April 2026

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Quick Start: Your First Grade in 3 Steps

Get up and running with GradeOS Simulator in minutes. This section walks you through starting a basic grading job from scratch.

Step 1: Power On the Machine

1. Click the **Power On (□)** button in the top toolbar.
2. Select your machine type: **Dozer** (for pushing soil) or **Excavator** (for digging).
3. The 3D viewport shows your machine positioned on the terrain.

Step 2: Select a Design and Create a Pad

4. In the **Design Tools** section of the toolbar, click **Flat Pad (□)**.
5. Click two points on the terrain to define the area where you want a flat pad.
6. The simulator creates a flat design surface at the clicked elevation.

Step 3: Drive and Grade

7. Use **WASD** keys or arrow keys to drive your machine over the design area.
8. Watch the **Volume Calculator** panel on the right: it updates **Remaining Cut** and **Job Completion %** as you grade.
9. Once 100% is shown in the job completion bar, your design is complete!

TIP: Turn on the Cut/Fill (✓) overlay to visualize progress. Green areas are on grade; blue is over-cut.

1. What is GradeOS?

GradeOS is an RTK GPS machine control system for dozers and excavators that costs approximately 10 times less than commercial solutions like Trimble and Topcon. The GradeOS Simulator demonstrates how the system works in a full-featured 3D terrain grading environment.

This simulator is designed for:

- Product demonstrations to investors
- Operator training on GPS-guided grading
- Design validation before field work
- Volume and cost estimation for earthwork projects

2. Interface Overview

The GradeOS Simulator interface is organized into five main areas:

2.1 Top Toolbar

All machine and design controls are grouped by category: Machine, Design, Terrain, Display, Features, and more. Each button has an emoji icon for quick visual recognition.

2.2 3D Viewport

The main work area showing your terrain in 3D perspective. Interact with it using:

- Scroll: Zoom in/out
- Right-drag: Orbit camera around terrain
- Middle-drag: Pan camera
- WASD/Arrow keys: Drive machine

2.3 Right Sidebar

Contains four key panels:

- Volume Calculator: real-time cut/fill volumes and cost estimates
- As-Built Tracking: survey coverage and accuracy
- Quick Demo Scenarios: one-click preset jobs
- GradeOS Scout: terrain mapping unit simulation

2.4 Status Bar

Located at the bottom of the viewport. Displays:

- Terrain name
- Design points and feature count
- Machine status (running/stopped)
- Cut volume this session
- Speed and heading

2.5 RTK Status Display

Located in the top-right of the 3D viewport. Shows GPS accuracy metrics:

- Fix type (RTK FIXED, RTK FLOAT, DGPS)
- Accuracy (typically 7–12 mm for RTK FIXED)
- Satellite count
- HDOP (horizontal dilution of precision)
- Base station distance

3. Machine Controls

The Machine section of the toolbar contains controls for powering on, selecting type, and operating your grading equipment.

3.1 Power On (□)

Click the Power On button to start the machine. It will appear in the 3D viewport at a starting position.

3.2 Machine Type Toggle

Switch between two machine types:

- **Dozer**: pushes and moves soil; ideal for creating pads and slopes. Controls blade height and angle.
- **Excavator**: digs and removes soil; ideal for cutting areas and trenches. Press SPACE to dig.

3.3 Fleet (□)

Deploy a second autonomous machine to grade parallel strips automatically. Useful for demonstrating multi-machine coordination and distributed grading.

3.4 Auto-Steer (⚙)

Engage automatic steering. The machine drives north-south parallel strips automatically, following the design surface. You can still control blade depth.

3.5 Manual Drive

Use keyboard controls to drive manually:

- W: Move forward
- A: Turn left
- S: Move backward
- D: Turn right
- SPACE (Excavator only): Dig

4. Design Tools

Design tools allow you to define the target grading surface. Most tools work by clicking two points on the terrain to define an area. A tooltip will indicate when you are in design mode.

4.1 Flat Pad (□)

Creates a flat surface at the clicked elevation. Click two corner points to define the padded area. Perfect for house pads, parking areas, or leveled work zones.

4.2 Slope (◥)

Defines a single plane slope between two points. Click the high point, then the low point to set the slope direction and grade.

4.3 Crown (□)

Creates a crowned road profile with a raised centre line and slopes on both sides. Prevents water pooling on roads.

4.4 Align + Profile

Click alignment points to place a route, then open the Profile Editor to design the cross-section. Advanced tool for complex road and corridor designs.

4.5 Dam (□)

Designs a retention dam with a pond floor and batters (sloped sides). Click to define the dam outline and the pond floor elevation.

4.6 Best-Fit (□)

Automatically fits the lowest-cut grade plane to existing terrain. Minimizes earthwork volume by matching the natural slope of the land.

4.7 Batter (□)

Creates a slope transition. Click the crest (top edge), then click the fall direction. The simulator generates batter slopes at a specified H:V ratio (horizontal to vertical). Used for embankments and cut slopes.

4.8 Irrigation Bay (□)

Designs an irrigation bay with auto-balanced cut and fill at a drain grade. Perfect for agricultural leveling.

4.9 Profile (✂)

Click two points to display a cross-section chart. Shows ground elevation vs. design with cut/fill shading. Useful for visualizing grade transitions.

4.10 V-Offset and Balance

A slider in the sidebar shifts the entire design surface up or down. Use the Balance button to auto-zero the cut vs. fill volumes, minimizing haul distances.

5. Terrain Types

The simulator includes 11 different terrain types to practice on. Click the Terrain toolbar button to select one.

Terrain Type	Description
Paddock	Flat agricultural field
Hilly	Rolling hills with moderate slopes
Valley	V-shaped valley terrain
Dam Site	Terrain suitable for dam/retention pond
Embankment	Sloped embankment terrain
Cut Hill	Hill requiring significant cut
Irrig Bay	Irrigation bay terrain for water management
Drone DEM (□)	Terrain from drone-captured digital elevation model
Road Cut (□)	Terrain for road cut construction
House Pad (□)	Multi-pad residential development terrain

6. Terrain Modification

Modify terrain using these tools by clicking two corners to define an area:

- **Smooth (□):** Smooths terrain elevation in the selected area.
- **Raise (□):** Raises terrain uniformly.
- **Lower (□):** Lowers terrain uniformly.

7. Feature Tools

Place linear and point features on your terrain. Click to add points; double-click to finish a line feature.

- **Waypoint (□):** Mark navigation points.
- **Measure (□):** Click two points to measure distance and slope.
- **Bank (⌒):** Mark banking features (curves).
- **Channel (≋):** Mark drainage channels.
- **Waterway (⌣):** Mark streams and water flow paths.
- **Pivot (◎):** Mark circular pivot irrigation points.
- **Boundary (⬡):** Mark property or job boundaries.
- **Trees (□):** Mark trees and vegetation.

8. Display Overlays

Toggle visual overlays to analyze your design and track progress. All are independent and can be combined.

8.1 Cut/Fill (□)

Colour-codes the design surface. Blue indicates cut areas (remove soil), orange/red indicates fill areas (add soil), and green indicates on-grade (at design elevation).

8.2 Contours (□)

Displays contour lines at 0.25m and 1m intervals for quick elevation reading.

8.3 Survey (□)

Shows GPS survey coverage. Coverage fills as the machine passes through areas in Scout mode, indicating measurement certainty.

8.4 Split Screen (□)

Splits the view to show the Blade Guidance panel with real-time elevation control information (see section 9).

8.5 Cut Simulation (□)

The machine blade actually cuts and fills terrain as you drive, dynamically modifying the ground surface in real time.

8.6 Flow (□)

Displays D8 drainage flow paths, showing where water will naturally drain on the graded surface.

8.7 Steep (⚠)

Highlights slopes steeper than 30 degrees, useful for identifying unstable areas.

8.8 Tolerance (✓ Tol)

Shows grade tolerance overlay: green = within ±25mm of design, yellow/orange/red = off grade by increasing amounts.

8.9 Stakes (□)

Displays grade stakes at grid points. Blue stakes = cut required, orange stakes = fill required. Stake height indicates depth.

8.10 Grade Direction (↗)

Shows arrows indicating design surface slope direction and grade percentage.

8.11 Labels (□)

Displays elevation numbers on major contours for quick reference.

8.12 Daylight (□)

Shows a yellow line where the design surface meets the existing ground, marking the cut/fill boundary.

8.13 Pass Heat Map (□)

Colour heat map shows how many passes the machine has made: orange = 1 pass, yellow = 2, green = 3–4, bright green = 5+. Indicates coverage completeness.

8.14 Grade Strips (□)

Shows the optimal grading sequence plan, visualizing parallel strips the machine should drive.

8.15 Replay (□)

Animates a time-lapse of terrain transformation from original to current graded state, great for demos.

8.16 Ghost Overlay (□)

Shows the original terrain as a transparent overlay. Amber shading = planned cut zones, grey-blue = areas not yet cut.

9. Blade Guidance (Split Screen Mode)

The Split Screen (□) toggle opens a Blade Guidance panel showing real-time elevation control for the machine blade.

9.1 Three Vertical Bar Displays

The panel shows three bars representing blade tip positions:

- Tip A (Left)
- Centre
- Tip B (Right)

9.2 Reading the Bars

Each bar indicates the required blade movement:

- Bar moves **UP** = machine must cut (blade is above grade)
- Bar moves **DOWN** = machine must fill (blade is below grade)
- Bar centred = **ON GD** (within ±25mm tolerance)

9.3 Depth Display

Numbers in millimetres show exact depth above/below grade. "ON GD" appears when within tolerance.

9.4 Cross-Section (X-Sec)

Click the X-Sec button to open a real-time cross-section window showing the blade's current cross-section perpendicular to the machine's heading.

10. RTK Status Panel

The GPS accuracy panel in the top-right of the viewport is critical for understanding machine positioning accuracy.

10.1 Fix Types

Fix Type	Colour	Description
RTK FIXED	Green	Full accuracy, 7–12mm horizontal. Ideal for grade control.
RTK FLOAT	Amber	Degraded fix, ~30mm accuracy. Machine jitters; not ideal for precise grading.
DGPS	Yellow	Basic accuracy, ~400mm. Not suitable for grade control work.

10.2 Accuracy Metrics

- **Satellites:** Number of visible GPS satellites (7+ required for RTK)

- **HDOP:** Horizontal Dilution of Precision (lower is better; <2 is ideal)
- **H.Acc:** Horizontal accuracy in metres
- **Base Distance:** Distance to RTK base station (critical for RTK; beyond 50–60 km accuracy degrades)
- **Correction Age:** Freshness of RTK corrections (should be <10 seconds)

11. Volume Calculator

Located in the right sidebar. Tracks volumes, costs, and job progress in real time.

11.1 Volume Metrics

- **Remaining Cut (m^3):** Volume of soil still to be cut
- **Remaining Fill (m^3):** Volume of soil still to be filled
- **Net (m^3):** Cut minus fill (positive = overall cut job)
- **Design Area (m^2):** Total area of design
- **Avg Haul Distance (m):** Average distance soil is transported

11.2 This Session

- **Cut This Session (m^3):** Volume cut by the machine since start
- **Engine Hours:** Simulated operating hours
- **Fuel Burn (L):** Estimated fuel consumption
- **Fuel Cost:** Calculated at $1.85/L

11.3 Cost Estimation

- **Rate ($/$m^3$):** Operator-set cost per cubic metre
- **Swell Factor:** Soil expansion factor (typical 1.1–1.3 for moved soil)

11.4 Job Completion Bar

Displays percentage of designed cells within ±25mm tolerance. Reaching 100% means the job meets specification.

12. Quick Demo Scenarios

Five preset scenarios load terrain and design automatically. Perfect for quick demonstrations or learning different grading types.

10. **Irrigation Bay Grade (□):** Flat bay with 0.05% fall grade, auto-balanced cut/fill
11. **House Pad Development (□):** Three flat pads at different elevations on hillside
12. **Road Construction Cut (□):** Diagonal road corridor with 3% grade
13. **Dam / Retention Wall (□):** Pond floor + batter slopes to embankment crest
14. **Scout Farm Mapping (□):** Machine auto-drives parallel strips building terrain DEM

13. Camera Modes

Switch camera views from the camera panel in the top-right of the viewport.

13.1 3D (Standard)

Standard perspective orbit view. Right-drag to rotate around terrain.

13.2 Plan

Orthographic top-down view, ideal for design work and visualizing strip patterns.

13.3 Follow

Camera follows the machine from behind, showing the work as the operator would see it.

13.4 Cab

First-person operator view from inside the cab. Most immersive perspective.

13.5 X-Sec Toggle

Opens a real-time cross-section window showing the blade's cross-section perpendicular to heading.

14. Probe & Measurement

Interactive tools for analyzing terrain and design.

14.1 Elevation Probe

Double-click on terrain to see a floating info card displaying:

- E/N coordinates from site centre
- Ground elevation
- Design elevation
- Cut/fill depth in mm
- Local slope %
- Compass direction

14.2 Measure Tool

Click the Measure (□) button, then click two points to see horizontal distance, height difference, and slope percentage.

15. Pitch Card (Investor Demo)

Press P to toggle a live stats overlay for investor demonstrations. Shows:

- Cut and fill volumes
- Job completion %
- Engine hours
- Productivity (m^3/hr)
- 3-year total cost of ownership: GradeOS vs Trimble

16. Keyboard Shortcuts

Shortcut	Action
W	Move machine forward
A / D	Turn left / right
S	Move machine backward
SPACE	Excavator dig
Scroll	Zoom in / out
Right-drag	Orbit camera
Mid-drag	Pan camera
Ctrl+Z	Undo
Ctrl+Y	Redo
H	Show keyboard help
P	Toggle Pitch/Status card (investor demo)
F	Toggle Field / Operator Mode (clean full-screen operator view)
Double-click terrain	Elevation probe (E/N, cut/fill, slope)

17. Screenshots & Export

17.1 Screenshot (□)

Click to save a PNG of the current 3D view.

17.2 CSV Export (□)

Exports design surface as a spreadsheet (E, N, ground, design, cut/fill per grid cell). Useful for CAD or GIS software.

17.3 As-Built Survey (□)

Exports the graded terrain as a survey dataset after work is complete.

17.4 Save / Load (□ / □)

Save your design to browser storage for later work. Load a previous design by clicking Load.

17.5 Job Report (□)

Generates a comprehensive PDF report including volumes, costs, machine setup, and productivity metrics.

18. GradeOS Scout Panel

A sidebar panel that simulates the Scout tractor-mounted terrain mapping unit. In Scout Farm Mapping scenario, coverage builds as the machine drives.

18.1 Scout Metrics

- **Coverage %:** Percentage of site mapped
- **Point Count:** Number of elevation points collected
- **Elevation Range:** Min to max elevation in dataset
- **Mean RL (Reduced Level):** Average elevation

19. Tips & Best Practices

TIP 1: Use the Cut/Fill (✓) tool to see at a glance which areas are on grade. Toggle Coverage (□) to visualize progress. Green = on grade, blue = over-cut, yellow/red = under-graded.

TIP 2: Enable Blade Guidance (□) split screen to see exact cut/fill depths in real time. Perfect for learning GPS-guided grading controls.

TIP 3: Use the Best-Fit (□) tool to minimize earthwork volume on projects with flexibility in design elevation.

TIP 4: Click the Balance (⚖) button to auto-zero cut vs. fill volumes, minimizing material haul.

TIP 5: For better coverage, use Auto-Steer (⚙) to drive parallel strips automatically. Watch the Passes (□) heat map to see coverage patterns.

TIP 6: Check RTK Status panel: green ='RTK FIXED' is ideal for grading. Yellow/Amber = reduced accuracy; move closer to base station.

TIP 7: Use Flow (□) overlay to check drainage patterns before completing a design.

TIP 8: Double-click terrain to probe elevation details at any location, including cut/fill depths.

TIP 9: Use Replay (□) overlay to play back the entire grading sequence for investor demos or training.

TIP 10: Press P to show the Pitch Card overlay with cost-of-ownership comparison for investor presentations.

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20. Troubleshooting

Issue: Low GPS Accuracy (DGPS or FLOAT mode)

- Cause: RTK base station too far away or incorrect setup
- Solution: Move closer to base station or wait for satellite geometry to improve. Check HDOP and satellite count in RTK status.

Issue: Design Not Creating Properly

- Cause: Clicking outside terrain bounds or insufficient points
- Solution: Ensure both clicks are on visible terrain. Use Plan view (camera panel) for precise point placement.

Issue: Machine Not Moving

- Cause: Machine not powered on or stuck in terrain collision
- Solution: Click Power On (□). Use Terrain Smooth (□) to clear obstacles if machine is stuck.

Issue: Cut Simulation Not Working (□)

- Cause: Overlay toggle is off or blade is not contacting terrain
- Solution: Toggle Cut Sim (□) on. Ensure blade is lowered to touch terrain.

21. Field / Operator Mode

Field Mode transforms GradeOS into a clean, distraction-free operator display — just like the screen inside a real machine cab. All editing panels disappear. The 3D terrain fills the entire screen. Three profile windows appear on the right side, and a grade readout bar runs along the bottom. Press **F** or click the □ **Field** button in the toolbar to enter or exit Field Mode at any time.

21.1 Three Profile Windows (Right Panel)

The right side of the screen is divided into three stacked windows, each giving a different view of the terrain around the machine. These update live every frame as the machine moves.
Long Section (top window) — A longitudinal terrain profile running along the machine's heading. Shows 40 m ahead and 10 m behind the blade. The blue line is the existing ground surface. The amber dashed line is the design surface. A vertical white line marks the blade position. The label above the blade shows cut or fill in millimetres (e.g. CUT 85mm or FILL 30mm). Elevation is shown on the left axis.
Cross Section (middle window) — A perpendicular slice across the machine at the blade position, 12 m either side. Blue = ground, amber dashed = design. The centre line marks the machine centreline. Cross-slope percentage is displayed in the top right. Use this window to see whether the blade is cutting an even cross-fall.
Plan View (bottom window) — A top-down 60 m view centred on the machine. Cells are coloured using the standard cut/fill palette (blue = cut, green = on grade, orange/red = fill). A white arrow shows the machine position and heading direction. A 20 m scale bar is shown in the bottom left. North indicator is top right.

21.2 Grade HUD Bar (Bottom Strip)

The bottom of the screen shows a compact heads-up display with the seven most important numbers for an operator:
Centre mm — Blade centre height error. Positive = cut required, negative = fill required.
Tip A mm / Tip B mm — Left and right blade tip height errors. Monitor these to keep the blade level across its full width.
Long Grade — Current longitudinal grade in percent, measured from the design surface slope at the blade position.
X-Slope — Cross-slope in percent perpendicular to travel direction.
Elev m — Current blade elevation in metres.
km/h — Machine ground speed.
GPS Status — Current RTK fix type (RTK FIXED / RTK FLOAT / DGPS), colour-coded green, amber, or yellow.

21.3 Floating Toolbar and RTK Strip

A small floating toolbar appears in the top-left corner in Field Mode. It has three buttons: the hamburger icon (☰) to exit Field Mode, a machine power toggle (⚙ ON/OFF), and an Auto-Steer toggle. These are the only controls available during field operation — everything else is hidden to keep the operator focused on the terrain.

An RTK status strip appears in the top-right corner of the viewport showing a colour-coded dot and the current fix type and accuracy string, mirroring the sidebar RTK display from the normal editor view.

21.4 Typical Field Mode Workflow

1. Set up your design in the normal editor view (select terrain, place pad or slope design, confirm volumes).
2. Start the machine and confirm RTK FIXED status.
3. Press **F** to enter Field Mode. The screen clears and the three profile windows appear.
4. Drive the machine. Watch the Long Section to plan your next pass. Watch the Cross Section to keep the blade level. Use the HUD bar for precise cut/fill numbers at the blade.
5. Press **F** again or click EXIT in the HUD bar to return to the full editor and check volumes or adjust the design.

Conclusion

The GradeOS Simulator v0.7 provides a complete, interactive environment for learning and demonstrating GPS-guided earthwork equipment. From basic flat pad grading to complex dam designs, the simulator supports a wide range of real-world scenarios.

Key takeaways:

- Start with Quick Demo Scenarios to learn interface basics
- Combine multiple display overlays for comprehensive grading feedback
- Use Blade Guidance split screen to master GPS-guided control
- Monitor RTK status for realistic accuracy understanding
- Generate reports and exports for project documentation

For more information on GradeOS hardware, visit the GradeOS website or contact the development team.